



Business Information Systems

EIS Types and Architecture

Week 06 · Lecture & self-study companion
Undergraduate · first-year friendly



How this hour flows

Segment	What happens	Time
Warm-up hook	One question that wakes the room up	~5 min
Part A — Background	How these systems were born	~8 min
Part B — The pyramid	The four classic EIS types	~10 min
Part C — ERP	The shared-database backbone	~10 min
Part D — CRM	The 360-degree customer view	~9 min
Part E — Architecture	How the software is actually built	~9 min
Video + wrap-up	Watch together, then recap	~9 min

Every abbreviation is spelled out in brackets the first time it appears — e.g. EIS (Enterprise Information System).



What you'll be able to do

By the end of this lesson, in your own words, you should be able to:

- Say what an EIS (Enterprise Information System) is — and why a company pays for one.
- Name the four classic types of business system and how they stack up.
- Describe what ERP (Enterprise Resource Planning) and CRM (Customer Relationship Management) actually do.
- Explain a shared database — the secret ingredient that makes these systems worth the money.
- Sketch the three-tier architecture nearly every system is built on, and why we split software that way.



Lesson 1

EIS Types and Architecture

Basic EIS types · background · key components of ERP & CRM





Setting the scene: what is an EIS?

- An EIS (Enterprise Information System) is a large piece of software that helps a whole organisation collect, store, and use its information.
- "Enterprise" simply means the entire organisation — not one team or one laptop.
- Behind a single cup of coffee sit a dozen questions: enough beans? supplier paid? is this a regular? best-selling store?
- No one person holds all that. The business runs on information — and software organises it.

That software is what this whole course is about.



Two words people mix up: IT vs IS

IT — Information Technology

- The gear and the plumbing.
- Computers, networks, servers, cables.
- Example: the cash register itself.

IS — Information System

- The whole arrangement.
- Technology PLUS people, data, and the steps they follow.
- Example: the register + cashier + price list + routine.



Background

How enterprise systems grew up — one practical headache at a time.

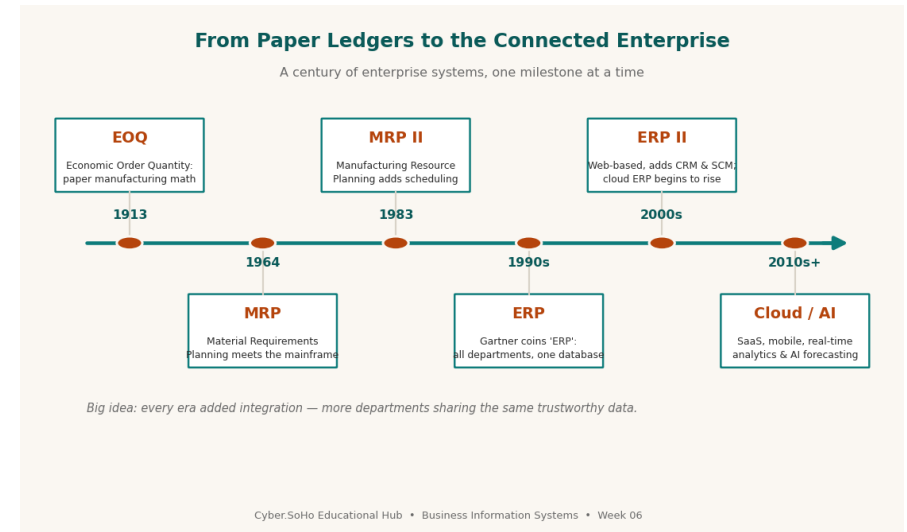


A How enterprise systems grew up

- Enterprise software wasn't invented all at once. It crept forward, each step fixing the last step's problem.

The whole story is really ONE idea, repeated: each generation let more of the business share the same trustworthy information.

- Watch the pattern: paper math → one factory job → the whole factory → the whole company → the whole supply chain → the cloud.
- More integration every single time. That's the heartbeat of this field.





The evolution story (1/2)

- 1913 — EOQ (Economic Order Quantity): a paper formula for the cheapest amount of stock to order. No computers — just smarter decisions with numbers.
- 1960s — MRP (Material Requirements Planning): turns a build plan into a parts list and schedule. "To build 500 bikes, I need which parts, and when?"
- 1980s — MRP II (Manufacturing Resource Planning): folds in machine time, labour, and cost — more of the factory plugged into one system.



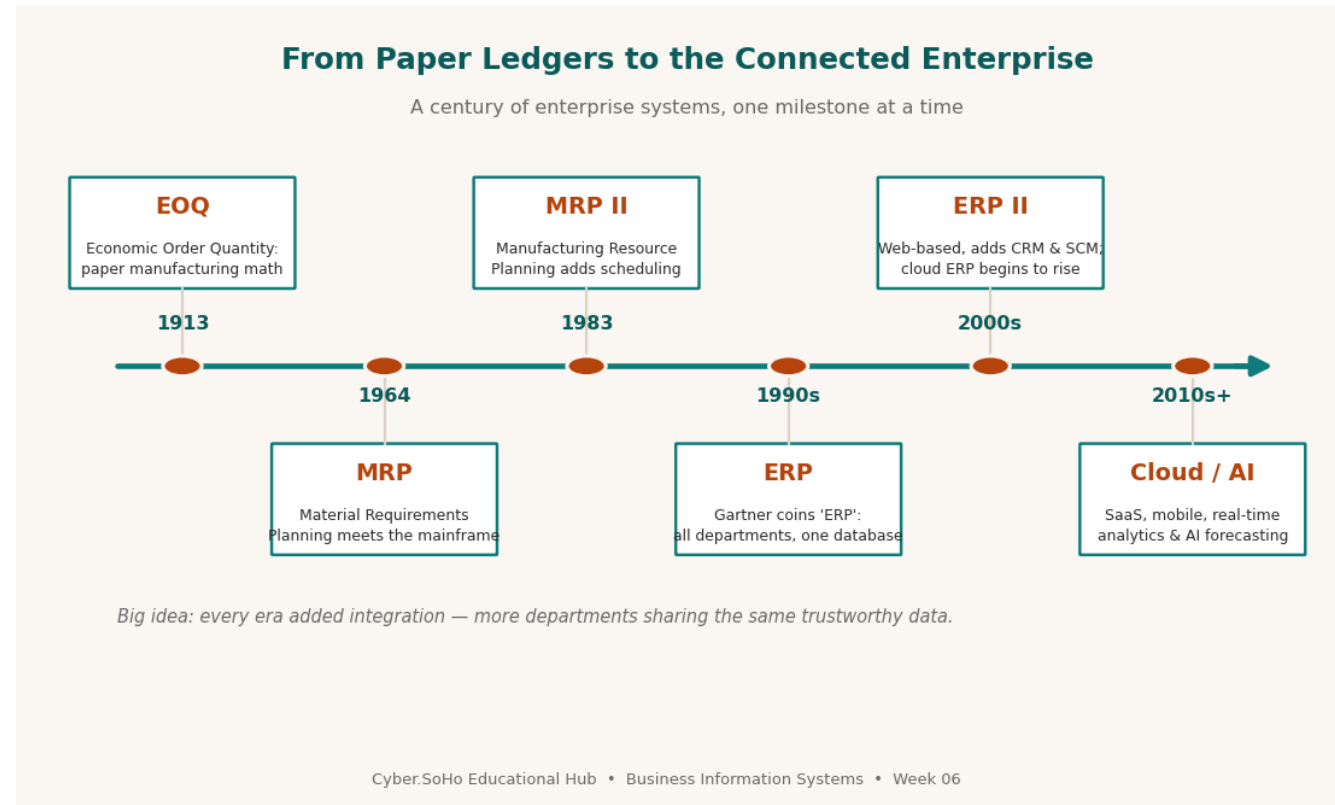
The evolution story (2/2)

- 1990s — ERP (Enterprise Resource Planning): the big leap. Gartner coined the term for software tying every department into one system + one database.
- 2000s — ERP II: connections reach outward to suppliers and customers.
- 2010s onward — Cloud + AI: software rented over the internet, with AI (Artificial Intelligence) for forecasting and automation.

Each era added more integration — so more people could trust the same data.



The evolution at a glance



From a 1913 paper formula to today's cloud + AI systems — every step added more integration.





Example: the three notebooks that never agreed

What this shows: A small shop outgrowing scattered, separate records.

Ties to the lecture: It's the exact pain that pushed companies from paper to MRP to ERP.

- Maya's Bikes keeps three notebooks: one for sales, one for parts, one for money. It works — barely.
- A customer asks for the blue tires. Maya checks parts: 4 left. She sells them — but forgets to cross them off.
- An hour later she promises the same 4 tires to someone else. Now she's oversold and her notebooks disagree.

The fix: store each fact ONCE, in a shared place everyone reads from. That single idea becomes ERP.



B

The information pyramid

The four classic EIS types — and how each sits on the one below it.



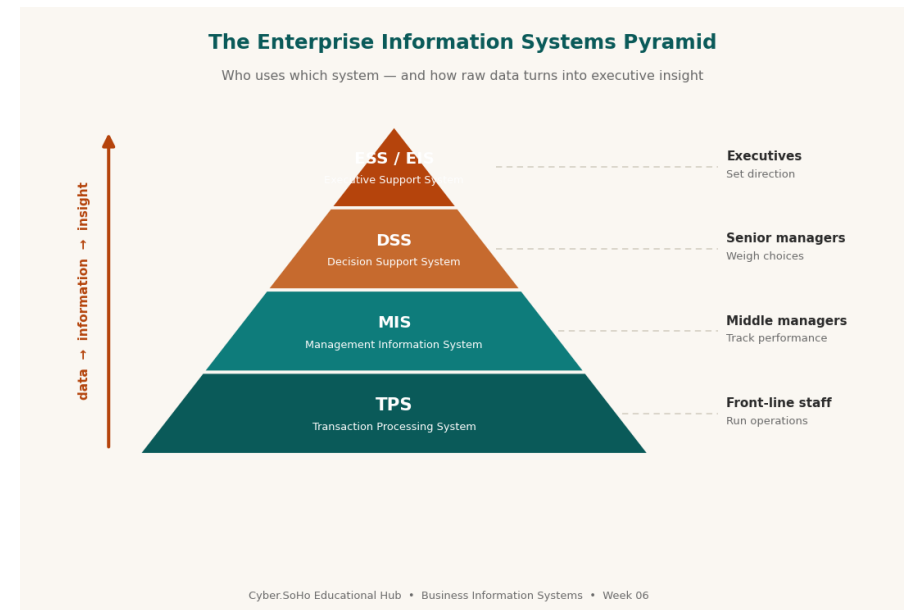
B

The basic EIS types

- Inside any organisation, systems stack up like a pyramid.
- At the bottom: a flood of tiny daily events. As you climb, those events become summaries, then decisions.

Four classic layers show up again and again: TPS, MIS, DSS, ESS.

- Golden rule: data flows UP and gets refined at every step.



B Layer 1 — TPS

TPS (Transaction Processing System) — the engine room.

- A transaction is one small business event: a sale, a payment, a booking, a clock-in.
- A TPS records these fast and accurately, thousands of times a day.
- Tap your card for a coffee → a TPS logs it.
- It doesn't analyse anything. It captures the raw facts, perfectly, every time.

B Layer 2 — MIS

MIS (Management Information System) — the summariser.

- Nobody wants to read 4,000 individual sales.
- An MIS takes the pile of transactions and turns them into reports a manager can skim.
- "Today's total sales were \$3,200; lattes were the top seller."
- It answers: what happened? Routine, structured summaries.

B Layer 3 — DSS

DSS (Decision Support System) — the helper for choices.

- It takes the summarised information and mixes in extra knowledge or "what-if" modelling.
- "If we drop cappuccino prices 10%, what happens to profit?"
- It supports a decision rather than just reporting the past.



Layer 4 — ESS

ESS (Executive Support System) — the high-level dashboard.

- Sometimes called an EIS (Executive Information System) — confusingly, same letters as our course title.
- Gives senior leaders a few big trends, comparisons with rivals, and the long view.
- It answers: where is this whole organisation heading?
- Leans on outside market data, not just internal numbers.

B The golden rule — and a naming caution

The golden rule

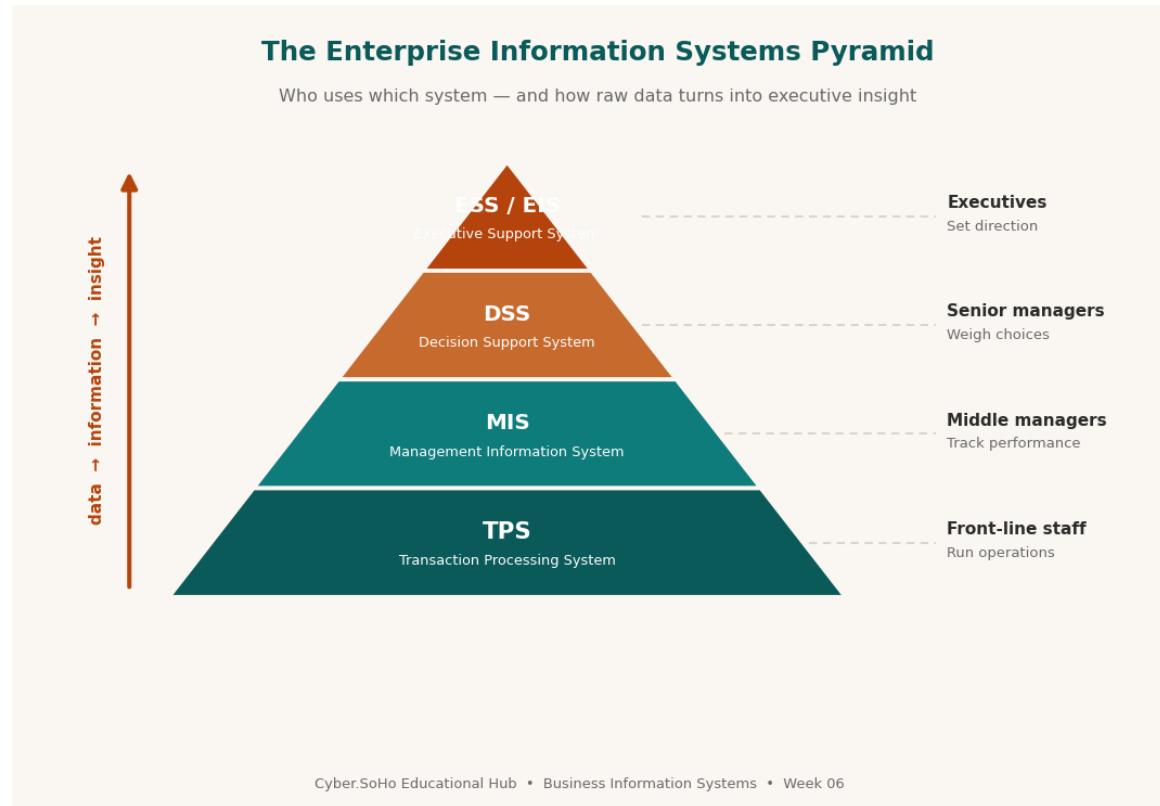
- Data flows UP and is refined at each step.
- Raw facts → information → decisions.
- Fewer people, bigger picture, longer horizon as you rise.

Watch the letters "EIS"

- Our title = Enterprise Information System (the umbrella term).
- Some books = Executive Information System (the top layer).
- When unsure, ask: enterprise-wide, or just the executives?

B

The information pyramid



TPS at the base feeds MIS, DSS, and ESS — data turns into insight as it climbs.

B

Example: one coffee sale climbs the pyramid

What this shows: A single sale travelling up all four layers, changing form at each step.

Ties to the lecture: It makes the pyramid move — one piece of data refined as it rises.

- TPS — 8:02 a.m.: the till records "2 x Latte, \$9.00, 08:02." Just the facts.
- MIS — at closing: "142 drinks sold, \$612 revenue, lattes on top." Boiled down to a few lines.
- DSS — "a 10% morning latte discount could lift volume ~15% but trim margin; net: slightly positive."
- ESS — the regional director sees only: "Store #7 revenue up 4% this quarter, ahead of the city."

Same coffee, four views — each layer serves a different person and a different size of question.



Enterprise Resource Planning

ERP — the shared-database backbone that cures Maya's notebook problem.





What is ERP?

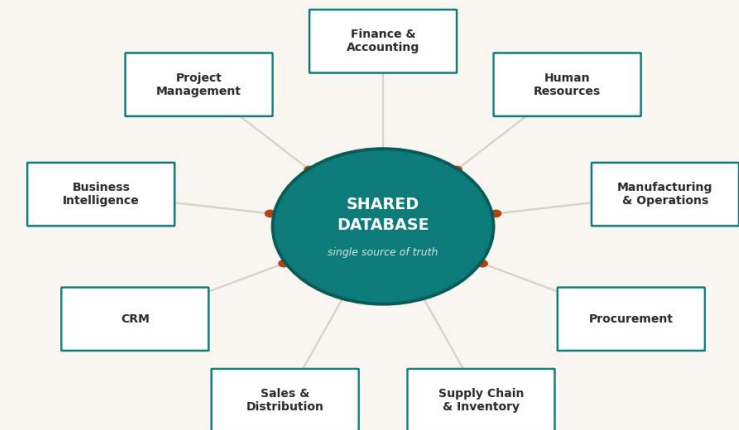
- ERP (Enterprise Resource Planning) ties a company's core departments into ONE integrated system built around ONE shared database.
- Finance, sales, inventory, purchasing, manufacturing, HR — all reading and writing the same central record.

The defining property: one shared database. Sell five mugs → inventory drops by five instantly, finance gains the revenue, reports update — with no re-typing.

- One fact, stored once, true everywhere.

ERP: Many Modules, One Shared Database

Every department writes to and reads from a single source of truth



Because the data lives in one place, a sale recorded by Sales is instantly visible to Inventory, Finance and BI — no re-typing, no conflicting copies.

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ERP components: modules (1/2)

- An ERP isn't one giant blob — it's a set of modules (apps that all plug into the same database).
- Finance & Accounting — the money: the general ledger, AP (Accounts Payable, money owed by the company), AR (Accounts Receivable, money owed to it).
- Inventory & Manufacturing — what's in the warehouse and what's being built.
- Sales & Order Management — quotes, orders, and invoices.



ERP components & properties (2/2)

- Procurement (Purchasing) — buying raw materials and supplies.
- HRM (Human Resource Management) — employees, payroll, time-off.
- BI (Business Intelligence) — dashboards that read across all the other modules.

Other key properties: best-practice workflows, real-time visibility, and scalability (start small, add modules as you grow).



Who makes ERP — and how big is it?

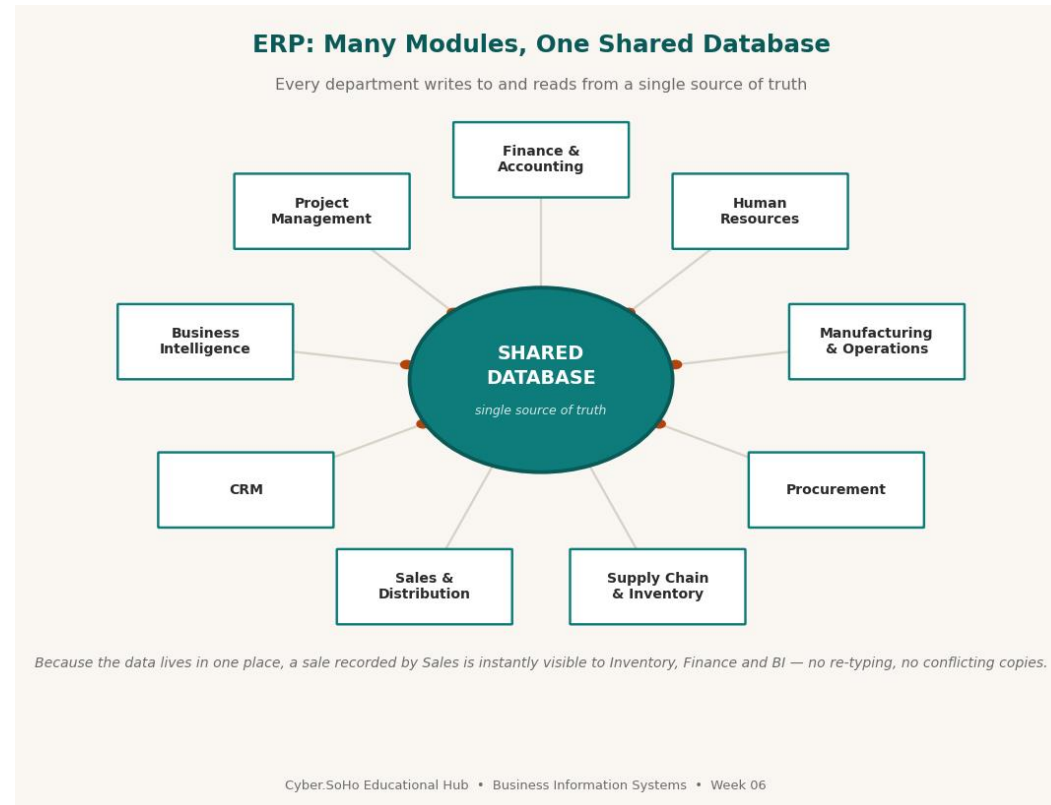
- The big names: SAP, Oracle (including NetSuite), and Microsoft Dynamics 365.
- For a vendor-neutral explainer, IBM keeps a readable overview.
- These systems run everything from corner shops to airlines.

The global ERP market is worth roughly \$80 billion a year — that's how central this idea has become.





ERP: one hub, many modules



Every module plugs into the same shared database — so the numbers can never disagree.

C Example: sell one mug, update the whole company

What this shows: The single shared-database idea — one action rippling correctly through several departments.

Ties to the lecture: It's the direct fix for Maya's notebooks and ERP's defining property in action.

- A customer buys 5 coffee mugs (\$60) from the online store. One click sets off:
- Sales records the order: 5 mugs, \$60.
- Inventory automatically drops the mug count by 5 — nobody updates it by hand.
- Finance automatically books \$60 of revenue. BI shows mugs climbing the best-seller list.

Four departments, one source of truth, no fact entered twice. That reliability is what ERP is selling.



Customer Relationship Management

CRM — the 360-degree view that looks outward at the people you sell to.

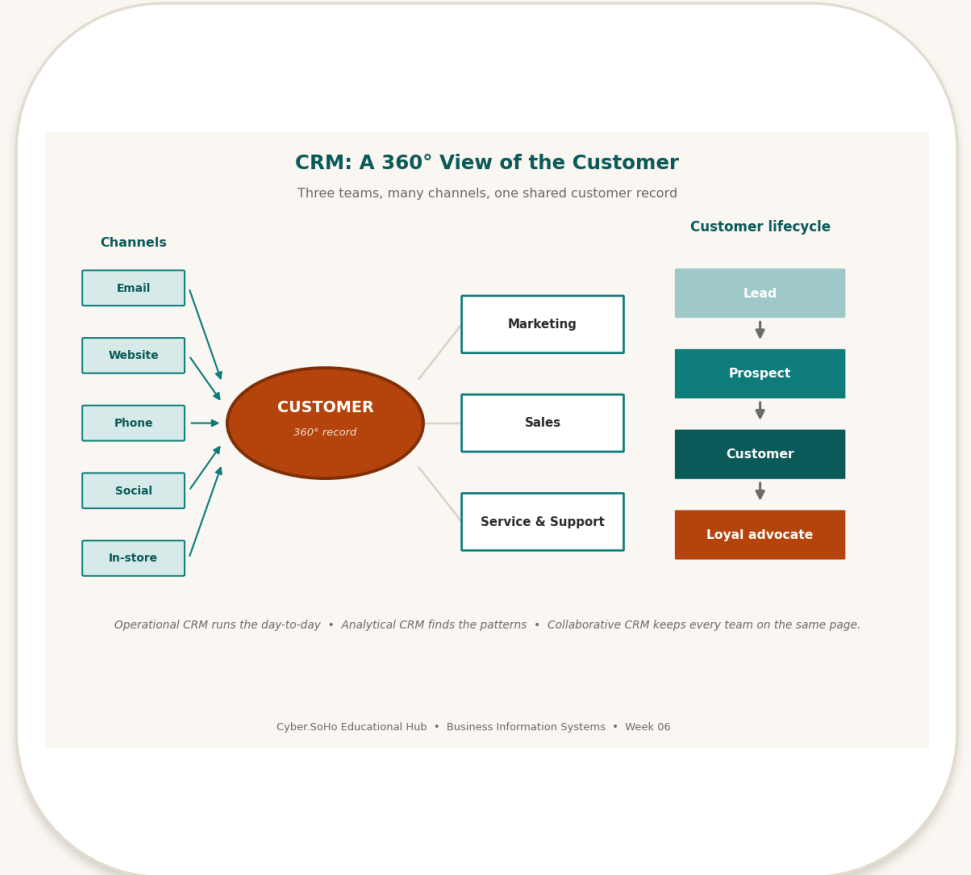


D What is CRM?

- If ERP looks inward at how a company runs, CRM (Customer Relationship Management) looks outward at the people it sells to.
- It tracks every customer and every interaction — calls, emails, purchases, support tickets — in one place.

The phrase to remember: the "360-degree view" — one complete, shared record per customer.

- Sales, marketing, and support all see the same full history. No more "let me transfer you."





The customer lifecycle

- CRM systems are built to move people through stages:
- Lead — a possible customer who's shown a flicker of interest.
- Prospect — a qualified lead worth pursuing.
- Customer — they bought something.
- Loyal advocate — a repeat customer who recommends you.

The CRM tracks where each person sits and helps nudge them to the next stage.



D The three flavours of CRM

- Operational CRM — the day-to-day machinery: managing contacts, logging sales, automating marketing and support.
- Analytical CRM — the brains: studying the data to spot your most valuable customers, or who's likely to leave (churn).
- Collaborative CRM — the connective tissue: sharing customer info across departments so everyone acts on the same knowledge.

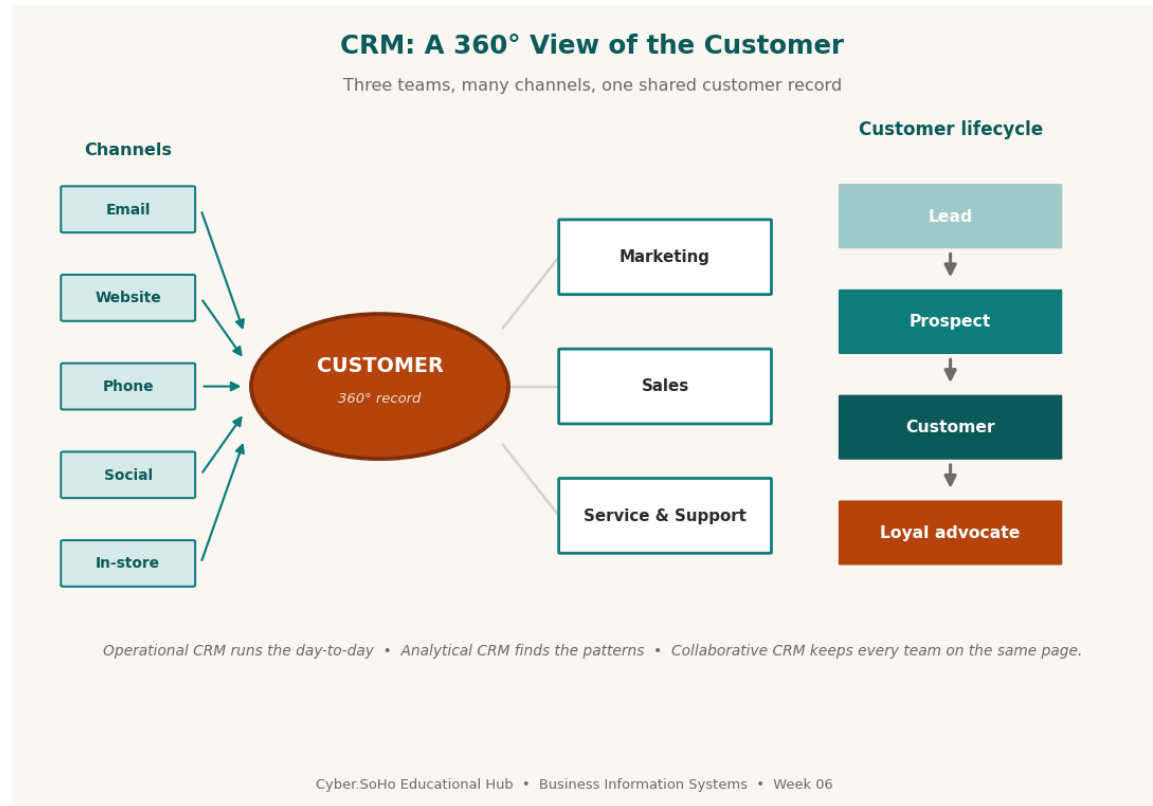
Together they turn scattered sticky notes into a shared memory of the customer.



Who makes CRM (and a cousin: SCM)

- The market leader is Salesforce, with HubSpot and Zoho also widely used — especially by smaller firms.
- Salesforce keeps a beginner-friendly explainer of what CRM is.
- One cousin you'll hear named alongside ERP and CRM: SCM (Supply Chain Management) — software for the flow of goods from supplier to customer.
- The comparison table later puts all three side by side.

D CRM: the 360-degree customer view



Many channels feed one shared customer record — visible to every team that needs it.

D Example: the customer who didn't repeat herself

What this shows: The 360-degree view in everyday life — one shared record that saves re-explaining.

Ties to the lecture: It's CRM's defining property and the lifecycle, seen from the customer's side.

- Priya buys a laptop online — she's now a Customer.
- Two days later it won't turn on. She emails support; the agent already sees her order, model, date, and a pre-sale note.
- He fixes it in one reply — no "what's your order number?"
- Later, marketing sees she has no accessories and sends a relevant laptop-bag offer — not spam. She buys, and tells a friend.

The single shared record is what makes the whole experience feel smooth.



Architecture

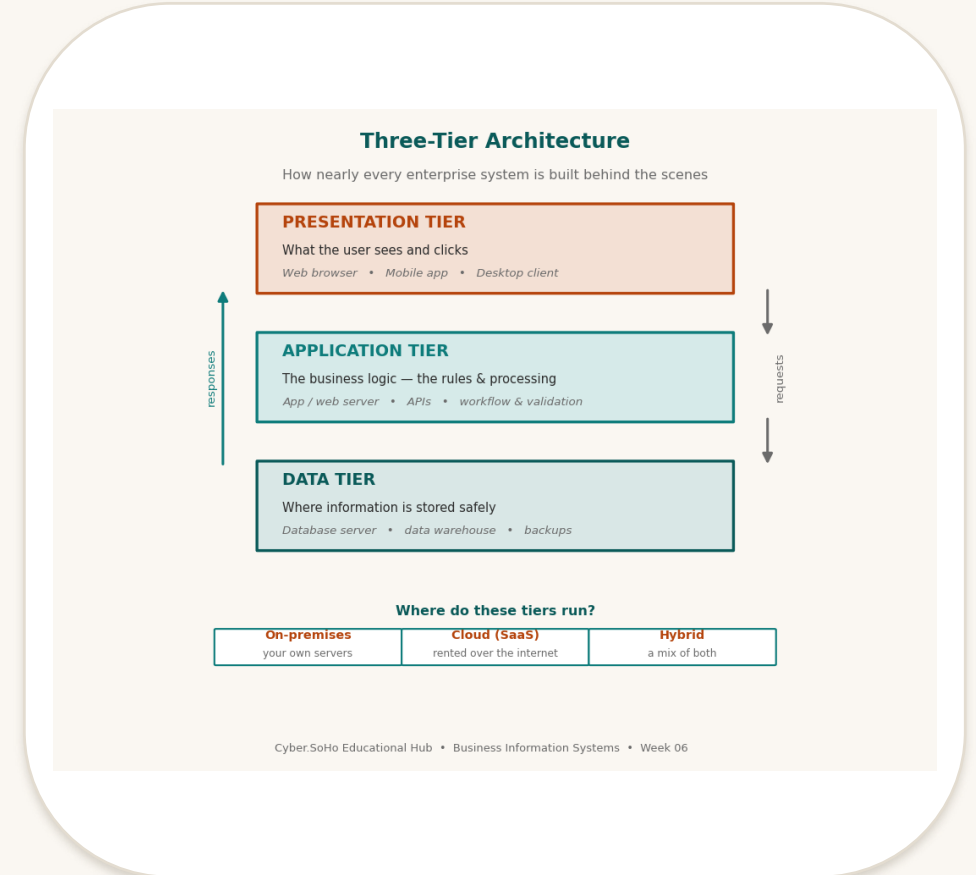
How these systems are actually built — tidiness and teamwork, not code.



E Three-tier architecture

- "Architecture" just means how the software is organised into parts. Almost every system splits into three tiers:
- Presentation tier — what you see and touch: screens, buttons, forms. The UI (User Interface).
- Application tier — the "brain": the business rules, e.g. "if stock < 100, suggest a re-order."
- Data tier — the DB (Database) where the facts live: customers, orders, inventory.

A click travels DOWN the tiers and the answer travels back UP to your screen.





How the pieces are packaged

- Monolithic — the whole app is one big block. Simple to start, but one change means rebuilding everything.
- Modular — divided into separate, swappable modules (remember ERP). Tidier and easier to extend.
- Microservices — broken into many small, independent services that each do one job and talk to each other.

Microservices are very flexible and scalable — how a lot of modern cloud software is built.



Where the system lives

- On-premises — the company runs the software on its OWN servers, in its own building. Maximum control, but you maintain it all.
- Cloud — it runs on a vendor's servers; you use it over the internet, often as SaaS (Software as a Service) rented monthly. Less control, far less hassle.
- Hybrid — a mix: some parts in-house (often sensitive data), some in the cloud.

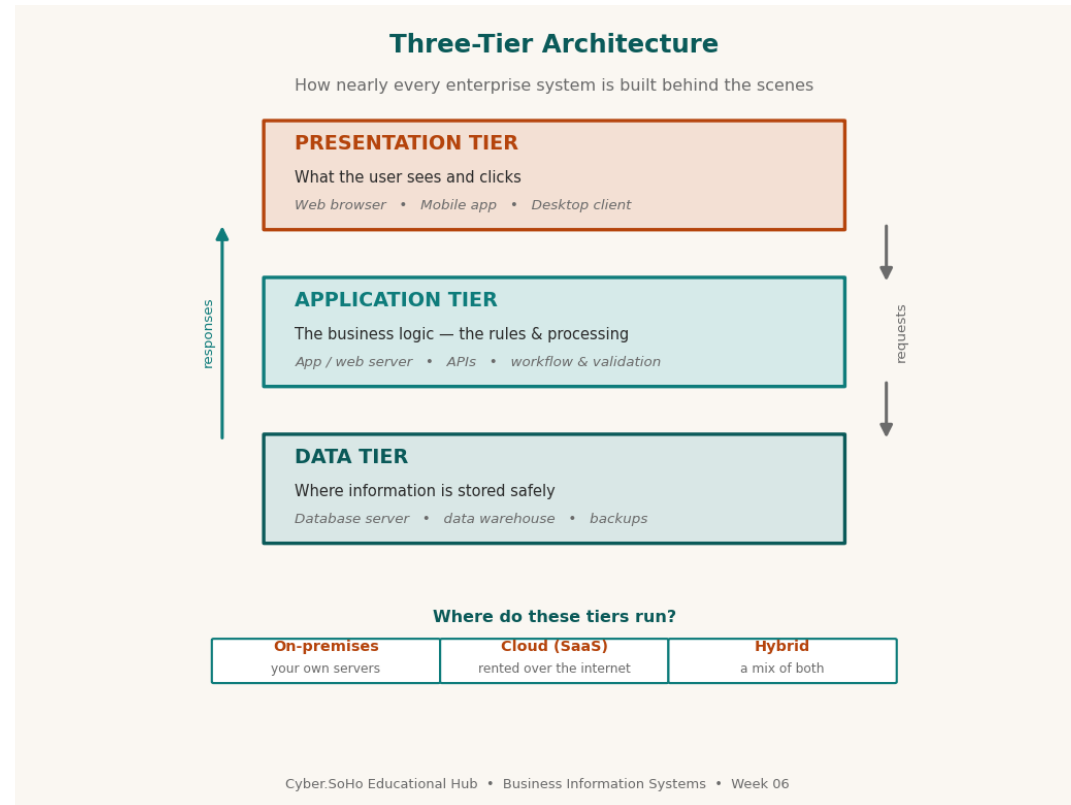


The glue: APIs & middleware

- Real companies rarely buy everything from one vendor — so systems must talk to each other.
- An API (Application Programming Interface) is an agreed set of rules that lets one program request something from another — like a waiter ordering from the kitchen via a fixed menu.
- Middleware is connector software that sits in the middle, helping different systems pass information back and forth.

Together they let a CRM, an ERP, and an online store behave like one joined-up whole.

Three-tier architecture



Presentation, application, and data tiers — separation is what keeps big systems safe and maintainable.

E Example: what happens when you press the button

What this shows: A single click travelling down the three tiers and back up.

Ties to the lecture: It makes "presentation / application / data" a path you can trace with your finger.

- You open your bank's website and click "Check balance."
- Presentation tier: the page registers the click and sends a request.
- Application tier: checks you're logged in and allowed, then decides it needs your balance.
- Data tier: the database looks up your account and returns the number — then it travels back UP to your screen.

All in a fraction of a second. Separate tiers are what keep it safe, fast, and maintainable at once.

Watch this — the lesson on video

► The video (under 10 min)

- "Types of Information Systems (TPS, MIS, and DSS)".
- A clear, beginner tour of Part B.
- youtube.com/watch?v=nQ4Q3iN7TMM
- Optional follow-up on CRM/ERP/SCM: [v=FZbG0EF0rW8](https://youtube.com/watch?v=FZbG0EF0rW8)

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ERP vs CRM vs SCM

Question	ERP	CRM	SCM
Main focus	Running the whole company	Managing customers	Moving goods supplier→customer
Looks mostly...	Inward (operations)	Outward (the customer)	Outward (suppliers & logistics)
Core idea	One shared database	One 360° record per customer	One view of the flow of goods
Typical users	Finance, HR, operations	Sales, marketing, support	Procurement, warehouse, logistics
Known vendor	SAP	Salesforce	Often an ERP/SCM module

Also provided as an editable spreadsheet: [ERP_vs_CRM_vs_SCM_Comparison.xlsx](#)



Glossary — the essentials

The four types

- TPS — Transaction Processing System
- MIS — Management Information System
- DSS — Decision Support System
- ESS — Executive Support System

The enterprise systems

- EIS — Enterprise Information System (umbrella)
- ERP — Enterprise Resource Planning
- CRM — Customer Relationship Management
- SCM — Supply Chain Management

Resources & links

- Gartner — coined "ERP" in the 1990s · gartner.com
- ERP vendors: SAP (sap.com), Oracle NetSuite (netsuite.com), Microsoft Dynamics 365.
- CRM vendors: Salesforce (salesforce.com), HubSpot (hubspot.com), Zoho (zoho.com).
- Plain-English explainers: IBM "What is ERP?" and Salesforce "What is CRM?"

Safety habit: scan any external link first at virustotal.com/gui/home/url.

- Full clickable list with descriptions is in the written lecture.



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